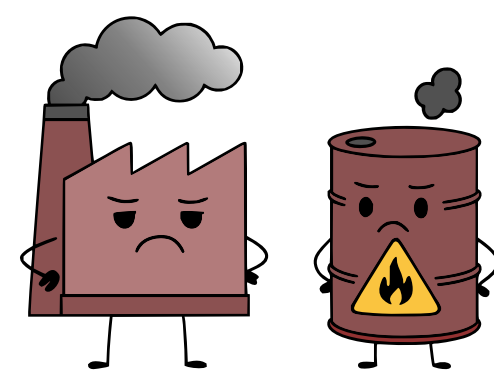


Probing the Dance of Interfacial Water in Photocatalytic Water Splitting with Phase- and Time-Resolved SFG and DFG

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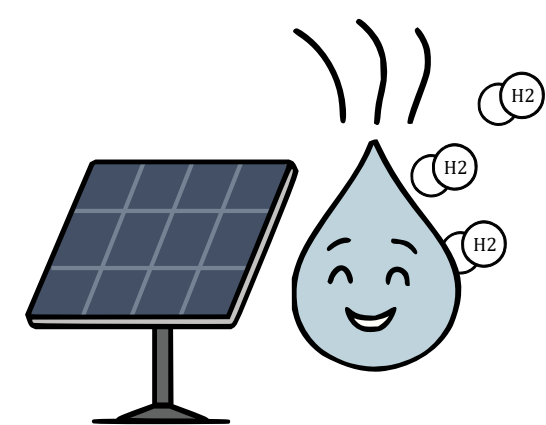
Introduction - Interfacial Water in Photocatalytic Water Splitting (PCWS)



Background

Why should we care about PCWS?

The direct conversion of sunlight and water into hydrogen fuel represents one of the most promising routes toward a carbon-neutral energy economy, placing photocatalytic water splitting (PCWS) at the centre of global decarbonisation strategies.



Despite more than 50 years of impressive research, we still don't have a scalable, commercially viable solution. While most efforts have focused on improving catalyst materials with band structure engineering, defect control, and co-catalyst designs, recent evidence points to a different bottleneck:

Problem: The reaction kinetics are heavily governed by the structure and organisation of water molecules at the catalyst surface.

At the nanoscale interface, water molecules undergo a vast cohort of non-equilibrium dynamics on ultrafast timescales: **hydrogen-bond network rearrangement, proton shuttling, and dynamic reorientation** under charge carrier generation and photo-induced electric fields.

We know they happen, but the specific dynamics of what and how for remain a mystery to us. These processes occur within the **first one or two molecular layers, on femtosecond to picosecond timescales**, and thus we have *very few techniques* that can simultaneously provide the information we need to understand them. A technique must provide:

We Need: Structural information with active-site specificity on ultrafast timescales... all in-situ during photocatalysis.

Molybdenum disulfide (MoS₂)

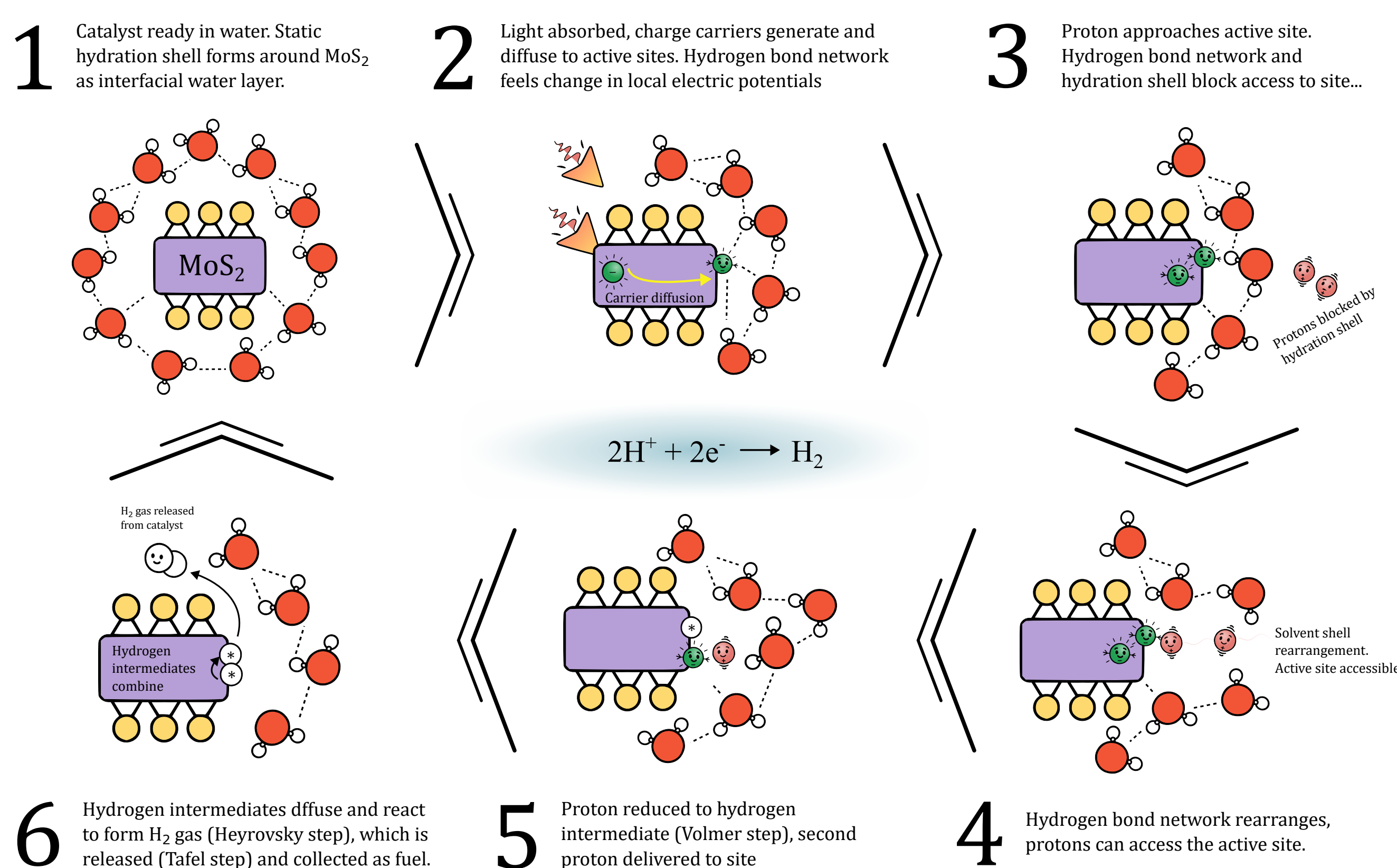
A promising **hydrogen-evolution reaction (HER)** catalyst for PCWS:

- It is earth-abundant and stable
- It has near-optimal hydrogen adsorption energy ($\Delta G_{H^*} \approx 0$)
- Its electronic structure can be tuned via defects/edges
- It integrates well with 2D heterostructures

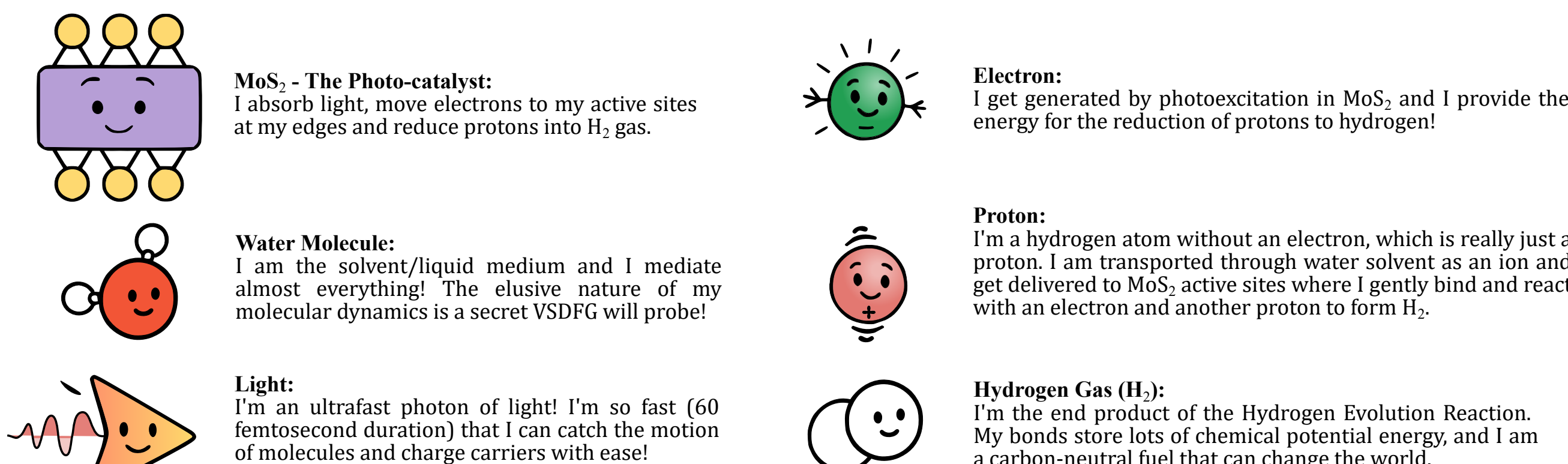
However, MoS₂ efficiency remains far below theoretical expectations...

The interfacial water layer must reorganise its H-bond network and reorient dipoles on pico- to nanosecond timescales to deliver protons to the active site. This motion is slow relative to the chemical kinetics, and thus limits efficiency.

Hydrogen Evolution Reaction (HER) in MoS₂



Meet the HER Team



The Technique: Vibrational Sum and Difference Frequency Generation Spectroscopy (VSDFG)

Interface-Specific, Structurally Sensitive, Ultrafast

Vibrational Sum and Difference Frequency Generation (VSDFG) uniquely probes only molecules without inversion symmetry, making it **inherently surface-specific** and perfect for investigating buried interfaces such as water on catalysts.¹

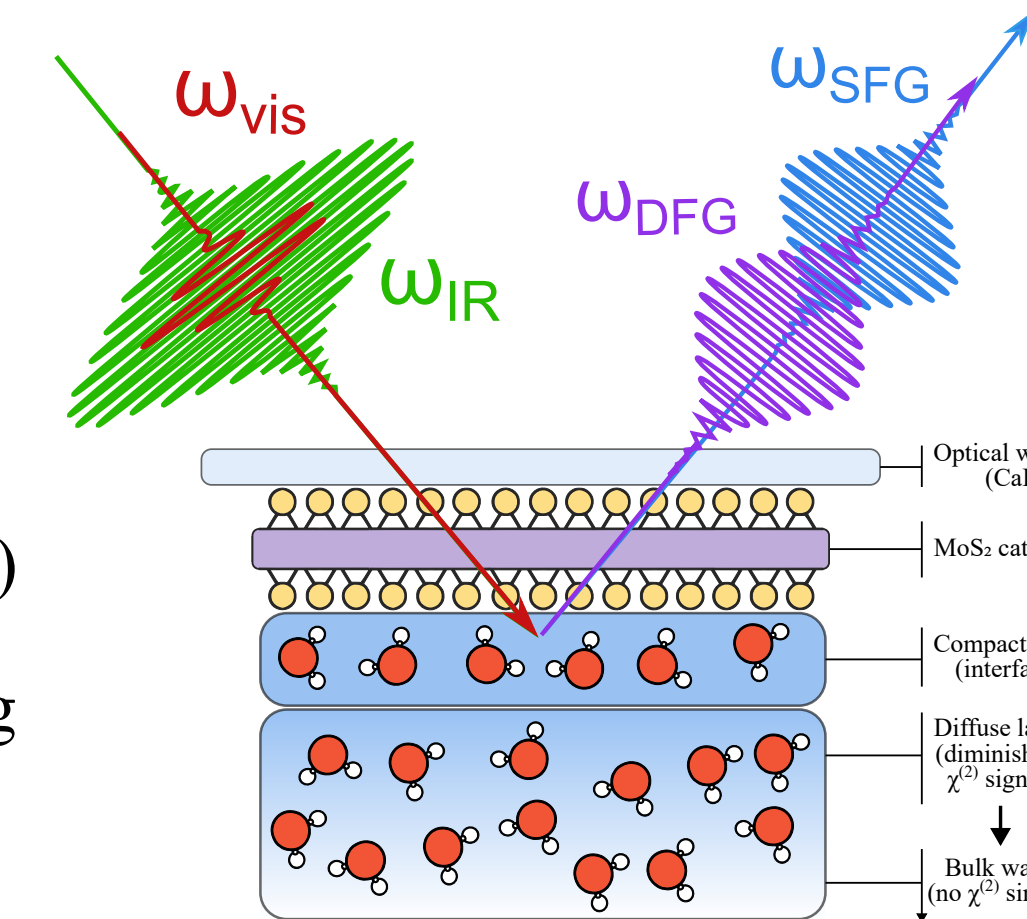


Figure 1: Diagram representation of nonlinear SFG signal generation. Only molecules near the surface layer can produce SFG signals due to broken inversion symmetry, conferring the surface specificity of the technique.¹

Key advantages:

- Sensitive to molecular orientation & bonding environment
- Operates on ultrafast timescales (femtosecond to picosecond)
- Can be used under realistic liquid-phase conditions
- Can be extended to pump-probe measurements for following reactions in real time

Heterodyne Detection:

A critical component, and challenge, for the work is the use of heterodyned detection, which uses interferometry to reconstruct both the amplitude AND the phase of the signals.² This enables:

- isolation of water resonances
- determination of water orientation (sign of $\chi_{lm}^{(2)}$)
- separation of $\chi^{(2)}$ vs $\chi^{(3)}$ contributions
- extraction of real dynamical changes during catalysis

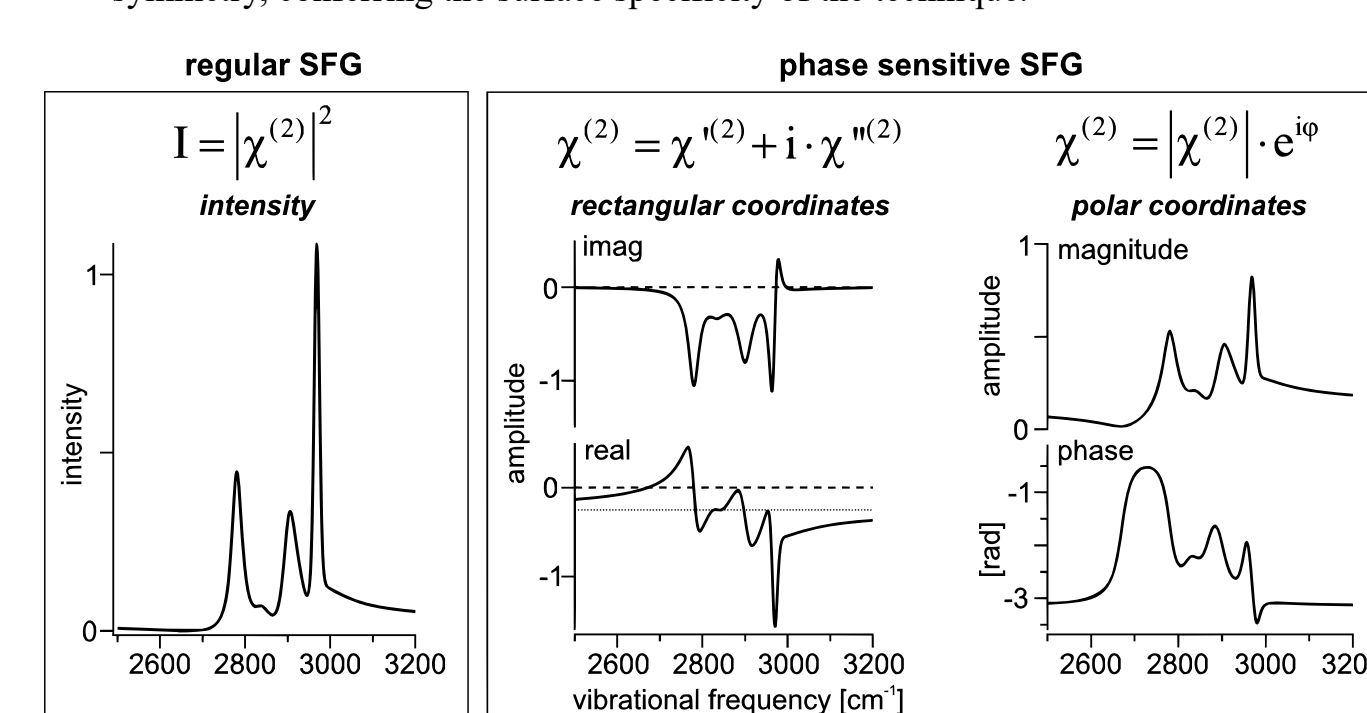


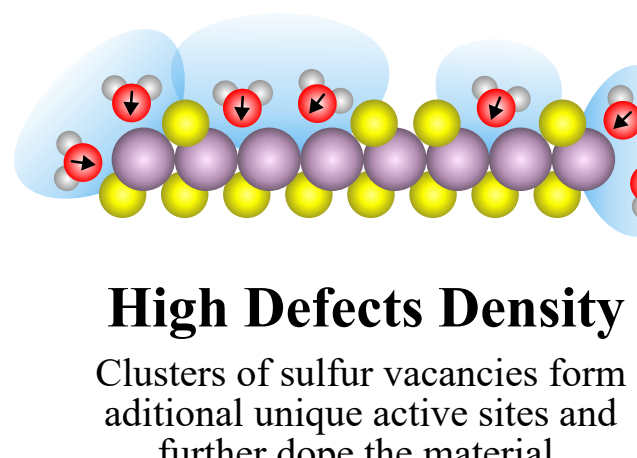
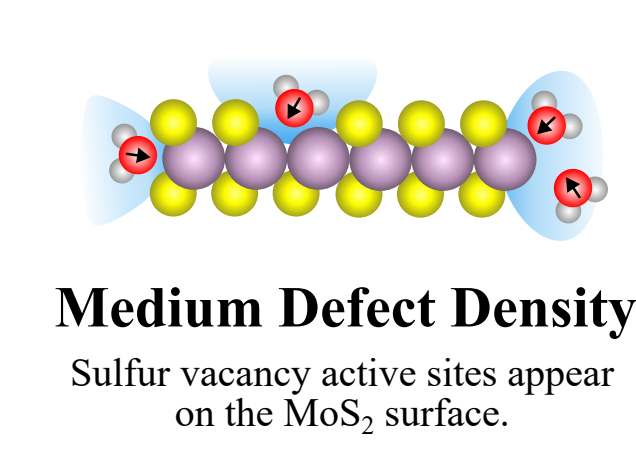
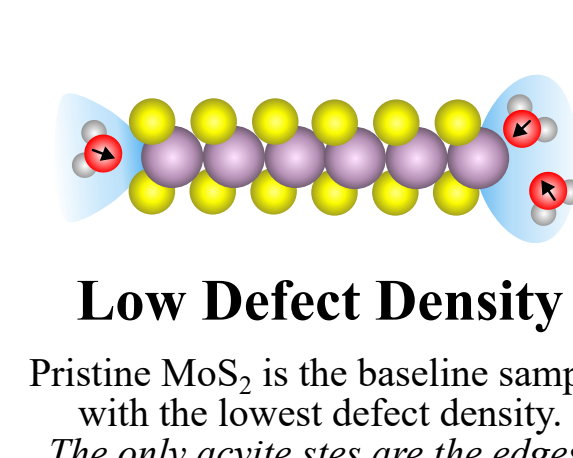
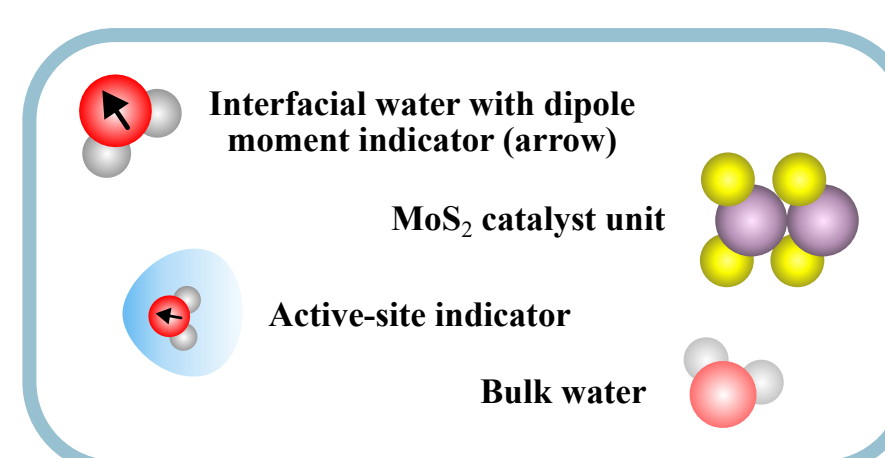
Figure 2: VSFG spectra for homodyne detection (regular SFG, left) and heterodyne detection (phase sensitive SFG, right). The extraction of phase generates a wealth of critical orientational information while simultaneously filtering out convoluting background signals.

This approach is crucial for measuring photocatalytic systems where transient photoinduced fields contribute complicated background signals.

Research Goals

1. Build a library of tunable MoS₂ materials

Defects alter the material's electronic properties via doping effects and create structurally and electronically distinct active sites to study



2. Quantify static interfacial water structure

Systematically perturb pH and temperature to follow changes to the thermodynamics

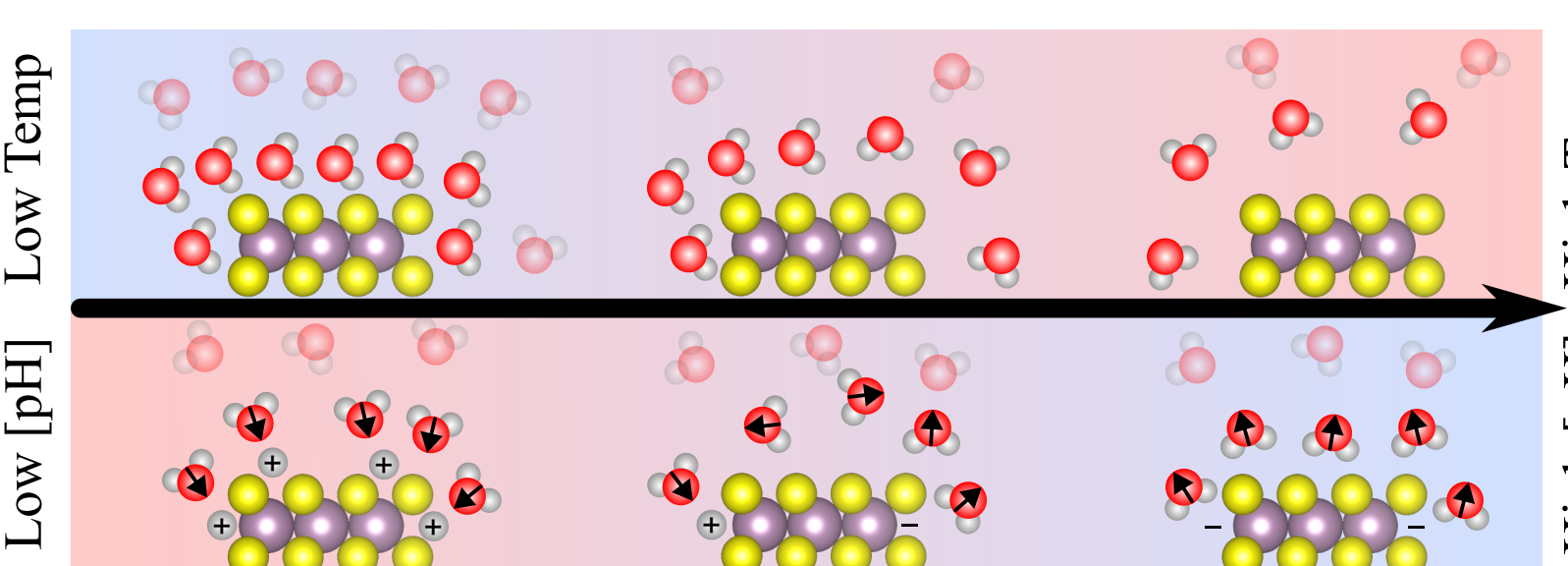


Figure 3: Representation of interfacial water structure variation as a result of varying temperature and pH.

Temperature measurements modulate thermal disorder

- Increasing thermal disorder weakens interfacial structure
- Creates shifts and broadening of OH vibrational frequencies, change in $\chi_{lm}^{(2)}$ sign

[pH] Measurements screen interfacial charges

- Water dipoles reorient: $\chi_{lm}^{(2)}$ sign flip
- Changes to H-bonded and free OH relative intensities

3. Measure ultrafast molecular motion during in-situ photocatalysis

Pump-induced photoexcitation to initiate photocatalysis, and time-resolved VSDFG to capture the structural evolution.

Pump-induced VSDFG follows the ultrafast dynamics:

- Charge-induced reorganisation of hydrogen bonding network
- Hydroxide ion formation and proton delivery
- Hydrogen desorption and hydrogen bond network reformation

Combining this analysis with ultrafast electronic characterisation, we will correlate structural reorganisation with the electronic drivers of photocatalysis in MoS₂ to fully elucidate the molecular dynamics of interfacial water throughout the HER.

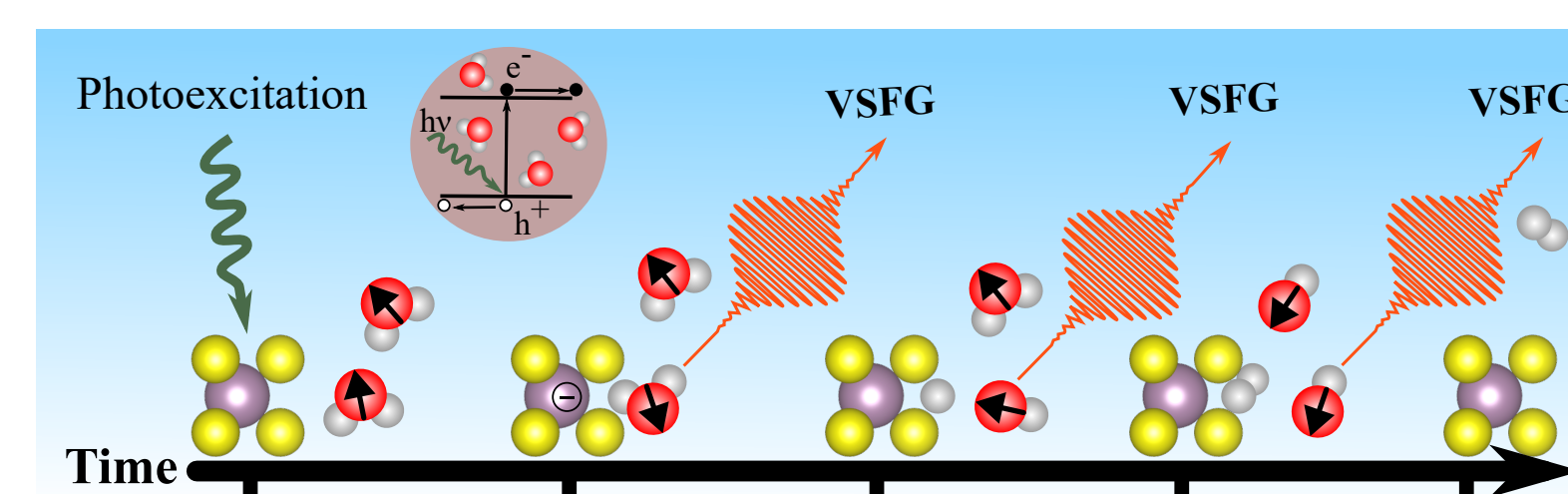


Figure 4: Representation of time-resolved photo-induced interfacial dynamics during HER.

Instrument: Optical Layout

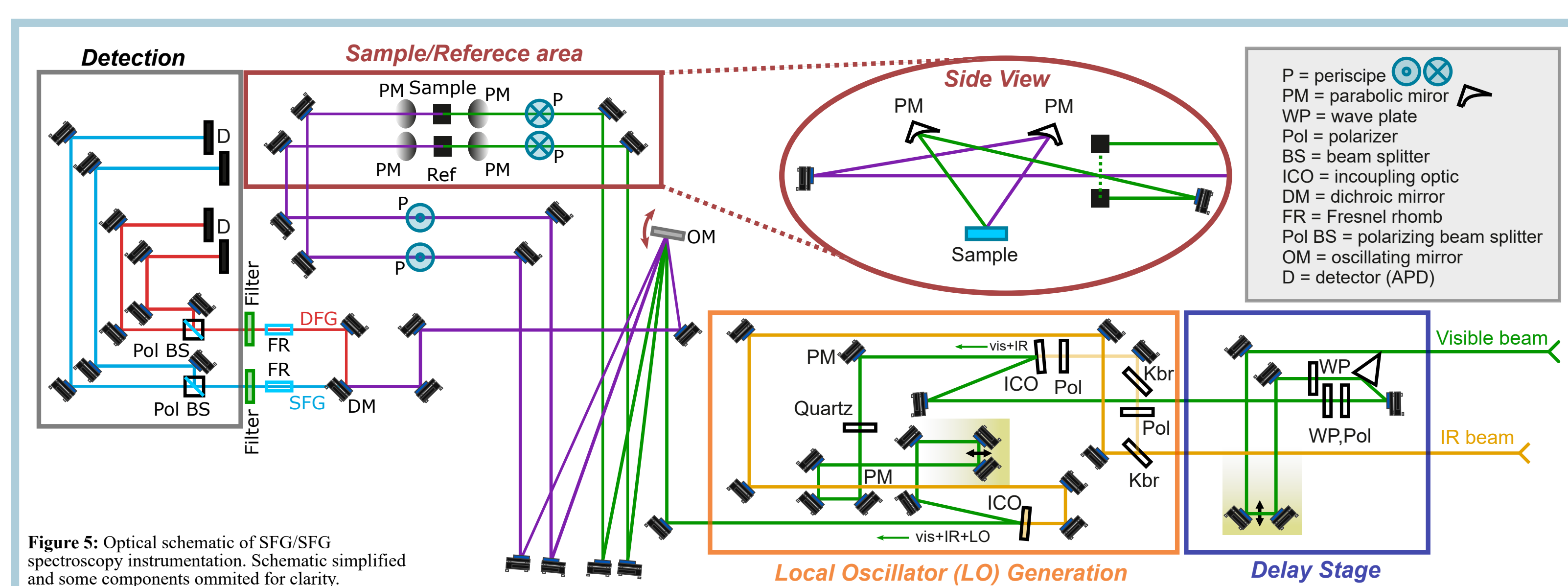


Figure 5: Optical schematic of SFG/SFG spectroscopy instrumentation. Schematic simplified and some components omitted for clarity.

Conclusions This research aims to yield the first molecular-level kinetic profile of water reorganisation at MoS₂ interfaces under HER-operational conditions, significantly advancing the understanding of HER interfacial processes in MoS₂. Combining SFG and DFG with robust electronic (Time-Domain Terahertz Spectroscopy) and physical characterisation, we will define which processes are immediate, persistent, or coupled to catalytically relevant fields, addressing a major limitation in the rational design of HER catalysts and establishing a transferrable analytical framework to fill a knowledge gap in and guide the design of future PCWS catalysts.

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